Investigation: BAHEMA EDUCAÇÃO S.A. (CVM code 701), fiscal year 2015, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Emp. Adm. Part. - Sem Setor Principal (origin: FCA); group: non-financial.
Before 2020, the company was called BAHEMA EDUCACAO S.A.
Before 2019, the company was called BAHEMA SA.
Trading codes (2023 FCA, version 2, filed on 2023-06-26): BAHI3 (Ações Ordinárias, Bolsa, since 2018-01-26).

Document used

2015 DFP, version 1, filed on 2016-03-22; origin: open data; type: individual; original scale: MIL.
Versions known on the date: 1 (2016-03-22).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=54269

Possible manipulation (engine v1, study 1)

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 11.383 (index)
Components: X1 = 0.927; X2 = 0.109; X3 = −0.073; X4 = 5.182 (indices)
Percentile in sector and year: 84 (n = 54)
The company's own history: median 10.636; Tukey range (1.5 × IQR): from 4.419 to 16.645; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 1/9 (7 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA: 0
- Cash above earnings (ACCRUAL): 0
- Change in leverage: 0
- Change in current ratio: 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin: —
- Change in asset turnover: 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
- No long-term debt in either year: the change in leverage is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Highlighted in the following year's comparative: 4 (threshold of R$0.4 million, 1% of total assets of the version used). Values in R$ million:

DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −4.7 in the version used (v1, filed on 2016-03-22); −0.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 3.9. Different value in the following year's comparative (restatement or reclassification).

BPP 2.02 Passivo Não Circulante: 3.8 in the version used (v1, filed on 2016-03-22); 2.4 in the next DFP's comparative (v1, filed on 2017-03-27); difference of −1.5. Different value in the following year's comparative (restatement or reclassification).

BPP 2.03 Patrimônio Líquido: 31.4 in the version used (v1, filed on 2016-03-22); 32.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 1.5. Different value in the following year's comparative (restatement or reclassification).

DRE 3.11 Lucro/Prejuízo do Período: −0.7 in the version used (v1, filed on 2016-03-22); 0.8 in the next DFP's comparative (v1, filed on 2017-03-27); difference of 1.5. Different value in the following year's comparative (restatement or reclassification).

Highlighted: differences in totals and key lines (total assets, current liabilities, non-current liabilities, equity, net revenue, net income and operating cash flow) above 1% of total assets of the version used. The other differences (reclassifications between lines and smaller differences) stay in the full list.

Full list: 46 accounts above the R$1,000 tolerance (group a, totals and key lines: 4; group b, reclassifications: 42), in the annex at the end.

What was known on the date

- DFP 2010, v1: 2011-03-04
- DFP 2011, v1: 2012-03-16
- DFP 2012, v1: 2013-04-01
- DFP 2013, v1: 2014-03-31
- DFP 2014, v1: 2015-03-28
- FCA 2015, v1: 2015-03-28
- FCA 2015, v2: 2015-05-04
- DFP 2015, v1: 2016-03-22
- DFP 2016, v1: 2017-03-27

DFP and FCA filed with the CVM by the reference date; ITR comes in a later phase.

Material facts and market notices filed by the date (IPE open data, since 2010): 51 material fact documents and 98 market notice documents.

Material facts filed in the 24 months before the date (from 2023-01-01 to 2024-12-31):

- 2024-11-14: subject omitted (possible person's name); see the document at the CVM
- 2024-09-30: subject omitted (possible person's name); see the document at the CVM
- 2023-10-30: Divulgação de Operação
- 2023-08-23: subject omitted (possible person's name); see the document at the CVM
- 2023-08-23: subject omitted (possible person's name); see the document at the CVM
- 2023-05-05: subject omitted (possible person's name); see the document at the CVM
- 2023-04-14: subject omitted (possible person's name); see the document at the CVM

Older material facts, the market notices and the link to each document are in the JSON and on screen.

Material facts and market notices filed with the CVM by the date, from the IPE open data (since 2010): filing date, category, subject (as filed, in Portuguese) and link. The content of the documents is not downloaded, read or summarized, and the list generates no alert, score or ranking. The counts are of documents: versions, resubmissions and English versions count separately; in the list, versions of the same document and documents with the same date, category and subject appear on a single line, with a link to each one.

Data quality

53 applicable checks; 3 failed.

- Cash flow closing balance = cash and cash equivalents on the balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

- Cash flow closing balance = cash and cash equivalents on the balance sheet (PENÚLTIMO): difference above the tolerance of R$2,000
- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: BDO RCS Auditores Independentes SS.
Auditor in the 2022 FCA: BDO RCS Auditores Independentes SS.
Auditor in the 2023 FCA: BDO RCS Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2015: Audilink & Cia. Auditores -> BKS Auditores. The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2016: BKS Auditores -> Grant Thornton Auditores Independentes Ltda. The outgoing firm audited 1 consecutive fiscal years (2015 to 2015), by the start and end dates in the FCA.
- 2019: Grant Thornton Auditores Independentes Ltda -> KPMG Auditores Independentes. The outgoing firm audited 3 consecutive fiscal years (2016 to 2018), by the start and end dates in the FCA.
- 2020: KPMG Auditores Independentes -> BDO RCS Auditores Independentes SS. The outgoing firm audited 1 consecutive fiscal years (2019 to 2019), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

Document excerpts (copied; no alert)

Excerpts copied from the documents, with the page, and the Reference Form table. It is not a signal family: it generates no alert, score or ranking. Absence of mention does not prove absence.
How the reader was tested: 20 held-out cases, a single measurement with the frozen reader (c99c953): precision 0.962, coverage 0.962, literal check on 132 of 132 excerpts, no natural-person name (docs/leitor_medida.json).

Explanatory notes · Related parties
The reader found no note with "partes relacionadas" (related parties) in the title.

Explanatory notes · Reverse factoring and forfaiting
The reader found no mention of reverse factoring (risco sacado), forfaiting or supplier agreements in the notes of this DFP. Absence of mention does not prove absence of the operation.

Auditor's report
Type of opinion: unqualified
- Excerpt omitted; see the auditor's report in the document

Reference Form
Form of 2016, version 6, filed on 2017-05-08.
It is the only version available in the open data; earlier versions are not there.
- Related party (name omitted) · text omitted (natural-person name) · Não há. · 29667380.98 · R$29.667.380,98

Limits of this section:

- In real documents, about 22% of the excerpts come out omitted by the name detector, including names of companies outside the registry.
- In tables, header lines (years, column titles) may come out as excerpts; the held-out cases had no tables with multi-line headers.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 36 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a F1); leitor-f1-1 (commit c99c953). The generation date is in the demo manifest (demo/manifesto.json).

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: BIOMA EDUCAÇÃO S.A.

Annex: full list of differences in the following year's comparative (R$ million)

(a) DFC (indirect method) 6.01 Caixa Líquido Atividades Operacionais: −4.7 in the version used, −0.8 in the comparative (difference of 3.9)
(a) BPP 2.02 Passivo Não Circulante: 3.8 in the version used, 2.4 in the comparative (difference of −1.5)
(a) BPP 2.03 Patrimônio Líquido: 31.4 in the version used, 32.8 in the comparative (difference of 1.5)
(a) DRE 3.11 Lucro/Prejuízo do Período: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) DFC (indirect method) 6.02.01 Adições de Investimentos: 6.2 in the version used, 0.0 in the comparative (difference of −6.2)
(b) DFC (indirect method) 6.01.01 Caixa Gerado nas Operações: −5.2 in the version used, 0.7 in the comparative (difference of 5.9)
(b) DFC (indirect method) 6.01.01.02 Despesas/(Receitas) Que Não Afetam o Caixa: −4.6 in the version used, −0.7 in the comparative (difference of 3.8)
(b) DRE 3.06.01.02 Outras Receitas Financeiras: 3.0 in the version used, 0.0 in the comparative (difference of −3.0)
(b) DFC (indirect method) 6.02 Caixa Líquido Atividades de Investimento: 6.7 in the version

used, 3.9 in the comparative (difference of −2.8)
(b) DFC (indirect method) 6.01.01.01 Lucro Líquido do Exercício: −0.7 in the version used, 1.4 in the comparative (difference of 2.1)
(b) DFC (indirect method) 6.01.02 Variações nos Ativos e Passivos: 0.5 in the version used, −1.4 in the comparative (difference of −2.0)
(b) BPP 2.02.02 Outras Obrigações: 3.4 in the version used, 1.9 in the comparative (difference of −1.5)
(b) BPP 2.02.02.02 Outros: 3.4 in the version used, 1.9 in the comparative (difference of −1.5)
(b) BPP 2.02.02.02.07 Dédicit Atuarial a Pagar: 2.0 in the version used, 0.5 in the comparative (difference of −1.5)
(b) BPP 2.03.05 Lucros/Prejuízos Acumulados: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) DRE 3.04 Despesas/Receitas Operacionais: −2.7 in the version used, −1.3 in the comparative (difference of 1.5)
(b) DRE 3.04.05 Outras Despesas Operacionais: −4.9 in the version used, −3.4 in the comparative (difference of 1.5)
(b) DRE 3.04.05.03 Provisão para Déficit Atuarial: −2.4 in the version used, −0.9 in the comparative (difference of 1.5)
(b) DRE 3.05 Resultado Antes do Resultado Financeiro e dos Tributos: −2.7 in the version used, −1.3 in the comparative (difference of 1.5)
(b) DRE 3.07 Resultado Antes dos Tributos sobre o Lucro: −0.1 in the version used, 1.4 in the comparative (difference of 1.5)
(b) DRE 3.09 Resultado Líquido das Operações Continuadas: −0.7 in the version used, 0.8 in the comparative (difference of 1.5)
(b) BPP 2.02.02.02.06 Outras Obrigações: 0.2 in the version used, 1.4 in the comparative (difference of 1.2)
(b) DFC (indirect method) 6.01.02.03 Variação no Passivo Circulante: 1.0 in the version used, −0.1 in the comparative (difference of −1.1)
(b) DFC (indirect method) 6.03 Caixa Líquido Atividades de Financiamento: −1.9 in the version used, −3.0 in the comparative (difference of −1.1)
(b) DFC (indirect method) 6.03.01 Dividendos e Juros s/ Capital Próprio: −1.9 in the version used, −3.0 in the comparative (difference of −1.1)
(b) BPA 1.01.01 Caixa e Equivalentes de Caixa: 0.0 in the version used, 0.7 in the comparative (difference of 0.7)
(b) BPA 1.01.03 Contas a Receber: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.01.03.02 Outras Contas a Receber: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPA 1.01.03.02.02 Outros Créditos: 0.7 in the version used, 0.0 in the comparative (difference of −0.7)
(b) BPP 2.02.02.02.03 Contribuição Social s/ o Lucro: 0.6 in the version used, 0.0 in the comparative (difference of −0.6)
(b) DFC (indirect method) 6.01.02.02 Variação no Ativo Não Circulante: 1.5 in the version used, 0.9 in the comparative (difference of −0.6)
(b) BPP 2.02.02.02.05 Parcelamento Federal-REFIS(Lei.12.996/14): 0.6 in the version used, 0.0 in the comparative (difference of −0.6)
(b) DFC (indirect method) 6.02.02 Ajustes de Avaliação Patrimonial em Controladas: 0.5 in the version used, 0.0 in the comparative (difference of −0.5)
(b) BPP 2.01.05.02.04 Outras Obrigações a Pagar: 0.4 in the version used, 0.0 in the comparative (difference of −0.4)
(b) DRE 3.04.04.02 Outras Receitas: 0.2 in the version used, 0.0 in the comparative (difference of −0.2)
(b) DRE 3.04.04.04 Reversão de Provisão: 2.6 in the version used, 2.9 in the comparative (difference of 0.2)
(b) DFC (indirect method) 6.01.02.04 Variação no Passivo Não Circulante: −1.5 in the version used, −1.8 in the comparative (difference of −0.2)

(b) BPP 2.01.01 Obrigações Sociais e Trabalhistas: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) BPP 2.01.01.02 Obrigações Trabalhistas: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) BPP 2.01.06 Provisões: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.06.01 Provisões Fiscais Previdenciárias Trabalhistas e Cíveis: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.06.01.03 Provisões para Benefícios a Empregados: 0.3 in the version used, 0.2 in the comparative (difference of −0.1)
(b) BPP 2.01.05.02.01 Dividendos e JCP a Pagar: 0.1 in the version used, 0.0 in the comparative (difference of −0.1)
(b) BPP 2.01.05.02.02 Dividendo Mínimo Obrigatório a Pagar: 0.0 in the version used, 0.1 in the comparative (difference of 0.1)
(b) DFC (indirect method) 6.03.04 Reversão Dividendo Prescrito e não Reclamado: 0.0 in the version used, 0.0 in the comparative (difference of −0.0)
(b) DFC (indirect method) 6.01.02.01 Variação no Ativo Circulante: −0.4 in the version used, −0.4 in the comparative (difference of −0.0)